

WOMEN TECHSTERS BOOTCAMP 5.1 • GROUP 18 CAPSTONE PROJECT

AI-Powered Small and Medium-Sized Enterprises (SMEs) Business Intelligence Dashboard

Simplifying Data Analytics for Better Business Decisions



Learning Tracks: Data Analysis & Technical Project Management

June – July 2026

What We'll Cover

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Project Overview & Problem

Why SMEs need this solution

2

The Dashboard Solution

Executive, Product & Customer views

3

Key Business Insights

What the data revealed

4

AI Integration

Turning dashboards into recommendations

5

Process, Challenges & Lessons

How we built it

6

Team & Next Steps

Who did what, and what's next

Turning SME Data Into Decisions

Small and Medium Enterprises drive jobs, innovation, and growth across Nigeria's economy. Every day they generate valuable data through sales, customer purchases, and financial activity, but most of it goes unused.

We built an AI-Powered SME Business Intelligence Dashboard that helps owners visualise sales performance, understand customers, monitor key metrics, and make informed decisions, no analytics background required.

BUILT FOR



SME Owners



Retail Business Owners



Entrepreneurs



Business Managers



Startups

Data-Rich, Insight-Poor



Data Sits Unused

Sales, customer, and project data is stored in spreadsheets or POS systems without ever being analysed.



No Time or Tools

Most SME owners aren't data analysts and lack the time, skills, or software to interpret their own numbers.



Decisions by Guesswork

Without evidence, decisions on sales, inventory, and growth rely on intuition rather than data.

This project closes that gap with a Business Intelligence dashboard that turns raw business data into clear, actionable visual insight for owners with no analytics background.

One Dashboard, Three Views



Executive Dashboard

- Total revenue & profit
- Number of projects
- Overall profit margin
- Monthly performance trend



Product & Service Analysis

- Top-performing services
- Profitability by category
- Residential vs commercial mix
- Cost vs revenue distribution



Customer & Regional Insights

- Geographic footprint by state
- Lead source performance
- Payment status matrix
- Repeat client tracking

Built With



Microsoft Excel

Data cleaning and preparation



Microsoft Power BI & Gemini

Dashboard development and visualisation



ChatGPT & Claude AI

Synthetic dataset generation, business insights and demo slide presentation



Microsoft Teams

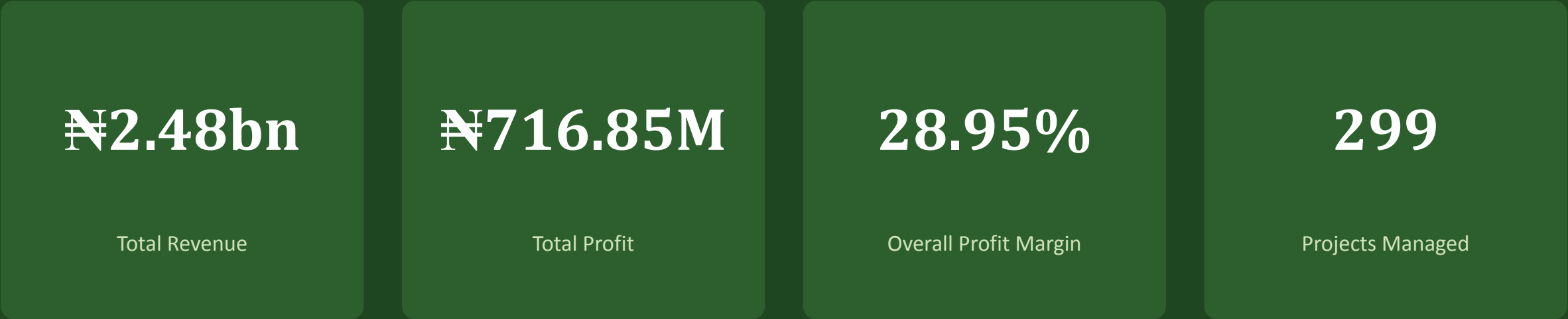
Team collaboration



GitHub

Project repository and version control

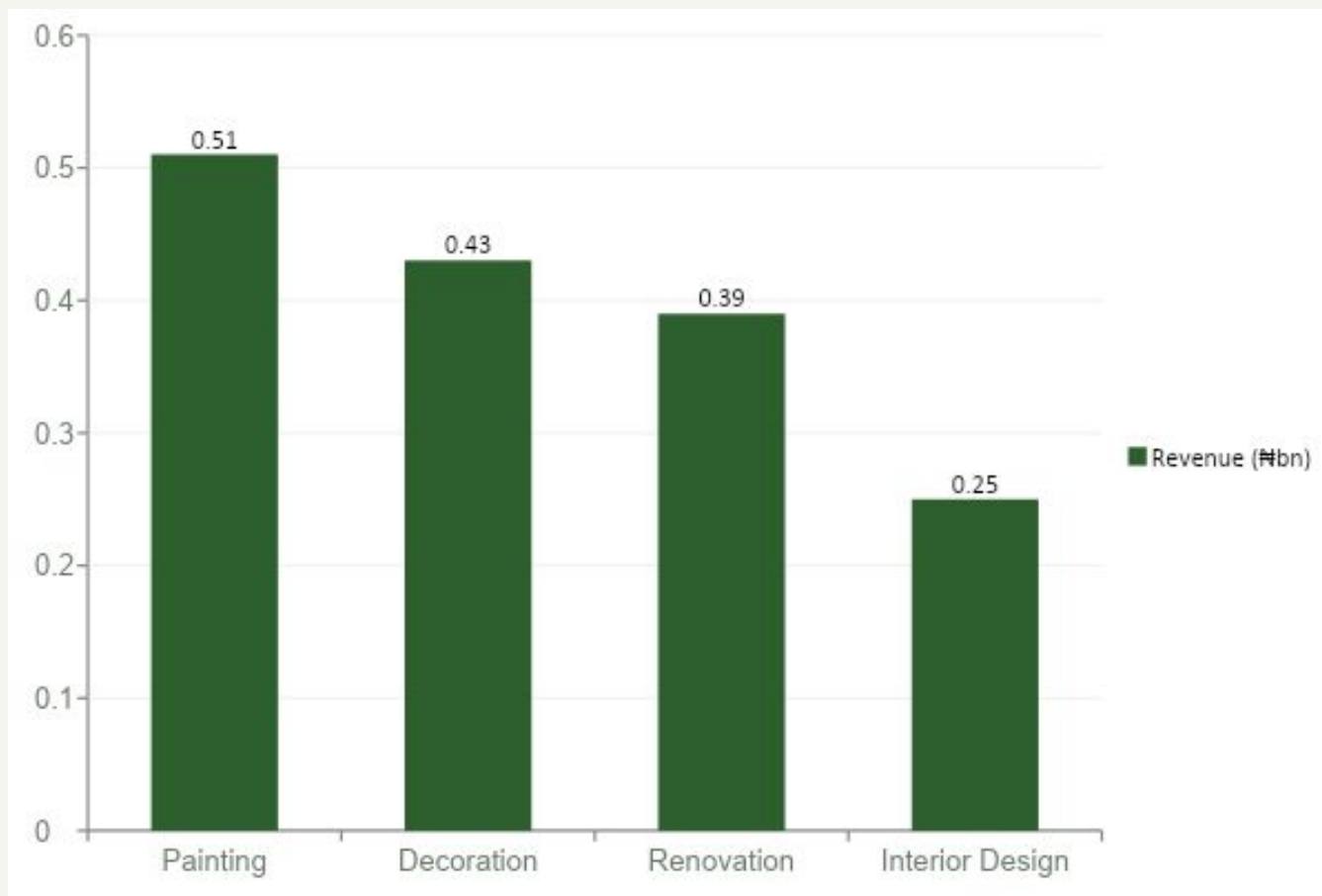
The Business, By The Numbers



The business converts revenue to profit efficiently



Painting Leads Revenue Generation

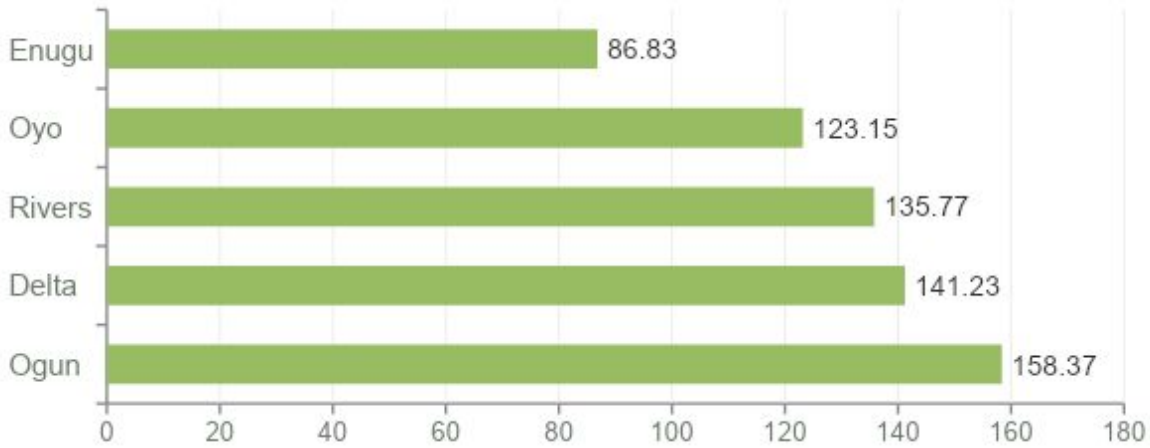


KEY TAKEAWAYS

- Painting, Decoration & Renovation are the strongest revenue drivers, together dominating the mix.
- Interior Design: the company's namesake service, is its lowest earner at ~~4~~0.25bn.
- Residential projects outperform Commercial by volume.

Where Clients Are, and Who Pays

TOP STATES BY REVENUE



LEAD SOURCES (BY VOLUME)

- Walk-in
- Facebook
- Website
- Referral
- Instagram

CLIENT PAYMENT MATRIX

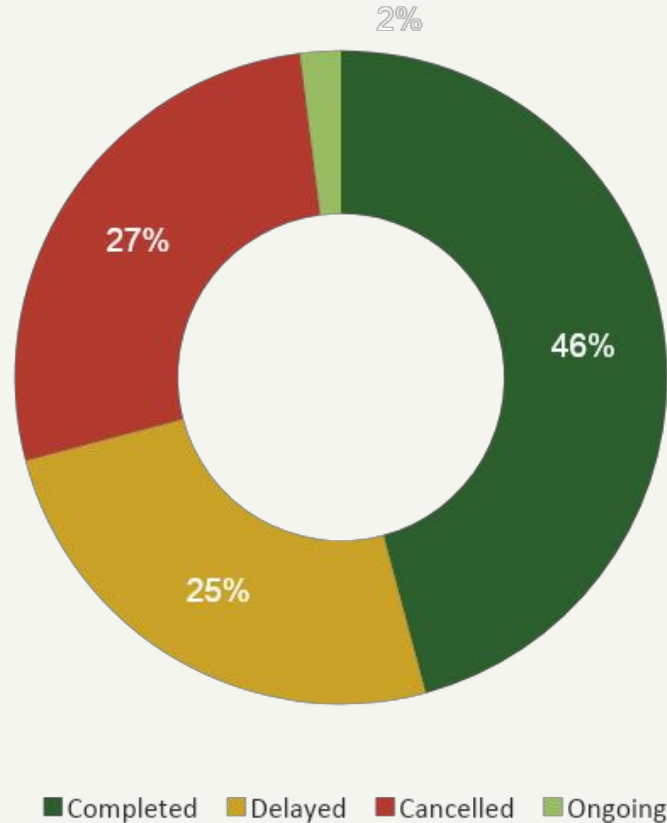
Client Type	Outstanding	Paid	Part Pay.
Individual	₦353.17M	₦398.21M	₦332.93M
Corporate	₦489.69M	₦427.15M	₦475.02M
Total	₦842.86M	₦825.35M	₦807.96M



Corporate clients hold the highest revenue AND the largest unpaid balance

₦489.69M in outstanding corporate debt creates a working-capital constraint — cash is tied up with the very clients driving the most revenue.

27% of Signed Projects Are Being Cancelled



81 Cancelled + 75 Delayed = 52.2% of the pipeline is not converting cleanly

Of 299 signed projects, only 137 (45.8%) reached completion, representing a severe leakage in the pipeline.

WHY IT MATTERS

High-volume lead channels (e.g. Facebook, Instagram) aren't yet cross checked against cancellation outcomes, marketing spend may be optimised for lead volume not lead quality.

From Dashboard Outputs to Business Recommendations

After data cleaning, EDA, and dashboard development, AI (Gemini) was applied to the dashboard outputs to generate an executive report translating charts into plain-language findings and action steps for non-technical SME owners.



1. Interpret Results

Read KPIs, trends and matrices from the dashboard



2. Identify Patterns

Surface trends, risks and growth opportunities



3. Flag Risk

Highlight cancellations, outstanding balances, delays



4. Recommend Action

Turn findings into practical, prioritised next steps

What the Data Is Telling Us

1

Profitable core, leaking pipeline

28.95% margin beats the informal-sector norm, but 52.2% of projects are delayed or cancelled.

2

Cash is trapped with your best clients

Corporate accounts drive the most revenue and hold the largest unpaid balance (N489.69M).

3

Revenue is concentrated outside the core brand

Painting, Decoration & Renovation outperform the company's namesake Interior Design service.

4

Lead volume isn't yet linked to lead quality

High-volume channels aren't cross-checked against cancellations a gap worth closing.

Data Quality, Challenges & Lessons Learned

DATA ISSUES FOUND & RESOLVED

- 14 records with unknown State values
- 67 “ongoing” projects had completion dates recorded
- 3 records with missing client names
- 81 cancelled projects had both start & completion dates validated before analysis

KEY CHALLENGES

- Sourcing a dataset with the right business variables
- Reconciling status vs. date inconsistencies
- Iterating DAX & layout for a clean, fast dashboard
- Coordinating across teams within a tight timeline

LESSONS LEARNED

- Insight quality depends on data quality, cleaning and validation come first.
- Synthetic datasets, built responsibly, can unlock realistic analysis when public data falls short.
- A good dashboard doesn't just display data it communicates a decision.
- AI + Power BI together turn analysis into action, faster than either alone.

Group 18

Role	Team Member(s)
Project Lead	Ayomide Elizabeth Temidayo
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Presentation Lead	Ayomide Elizabeth Temidayo, Morenikeji Bankole,
Quality Assurance	Ayomide Elizabeth Temidayo, Jadesola Fawibe, Washika Comfort Msurshima,

Sustainable Development Goals

SDG 8

Decent Work & Economic Growth

Supporting SMEs with data-driven decisions that improve productivity and business growth.

SDG 9

Industry, Innovation & Infrastructure

Promoting digital transformation through Business Intelligence and Artificial Intelligence.

CONCLUSION

This dashboard demonstrates how Business Intelligence and AI together can simplify decision-making for SMEs; turning raw data into visual insight and visual insight into action. Beyond the technical build, the project strengthened our skills in collaboration, analytics, and project management, and lays the foundation for a future self-service BI platform for SMEs.

Thank You

Questions & Discussion are Welcome



github.com/ChiedzaRene/WTB-Capstone-Project-Group-18-Smart-Analytics-For-SMEs-In-Nigeria-